

Initial Parameter Configuration.

- CONTEXT_LENGTH: 30
- D_MODEL: 32
- *_LAYERS: 2
- *_HEADS: 4
- DROPOUT: 0.1
- LEARNING_RATE: 1e-4
- BATCH_SIZE: 64
- EPOCHS: 20 (with PATIENCE=10)

DROPOUT: 0.1

Epoch 01/20 -> Train Loss: -1.26991204, Val Loss: -1.65727360
New best validation loss: -1.65727360. Saving model...

Epoch 02/20 -> Train Loss: -2.20737763, Val Loss: -2.21042946
New best validation loss: -2.21042946. Saving model...

Epoch 03/20 -> Train Loss: -2.50152380, Val Loss: -2.42937358
New best validation loss: -2.42937358. Saving model...

Epoch 04/20 -> Train Loss: -2.64044149, Val Loss: -2.35278372
Validation loss did not improve. Patience: 1/10

Epoch 05/20 -> Train Loss: -2.73635860, Val Loss: -2.59325069
New best validation loss: -2.59325069. Saving model...

Epoch 06/20 -> Train Loss: -2.79875934, Val Loss: -2.73347369
New best validation loss: -2.73347369. Saving model...

Epoch 07/20 -> Train Loss: -2.83933081, Val Loss: -2.61346611
Validation loss did not improve. Patience: 1/10

Epoch 08/20 -> Train Loss: -2.88090314, Val Loss: -2.80592543
New best validation loss: -2.80592543. Saving model...

Epoch 09/20 -> Train Loss: -2.90326997, Val Loss: -2.74745058
Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -2.93674289, Val Loss: -2.73024203
Validation loss did not improve. Patience: 2/10

Epoch 11/20 -> Train Loss: -2.95575961, Val Loss: -2.88569784
New best validation loss: -2.88569784. Saving model...

Epoch 12/20 -> Train Loss: -2.97721556, Val Loss: -2.72488405
Validation loss did not improve. Patience: 1/10

Epoch 13/20 -> Train Loss: -2.99184759, Val Loss: -2.90419203
New best validation loss: -2.90419203. Saving model...

Epoch 14/20 -> Train Loss: -3.01072159, Val Loss: -2.88882065
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -3.02630497, Val Loss: -2.94505787
New best validation loss: -2.94505787. Saving model...

Epoch 16/20 -> Train Loss: -3.05087575, Val Loss: -2.88568141
Validation loss did not improve. Patience: 1/10

Epoch 17/20 -> Train Loss: -3.05573125, Val Loss: -2.92449716
Validation loss did not improve. Patience: 2/10

Epoch 18/20 -> Train Loss: -3.06843116, Val Loss: -2.97391046
New best validation loss: -2.97391046. Saving model...

Epoch 19/20 -> Train Loss: -3.07660289, Val Loss: -3.06566454
New best validation loss: -3.06566454. Saving model...

Epoch 20/20 -> Train Loss: -3.08992422, Val Loss: -2.98136848
Validation loss did not improve. Patience: 1/10

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Final Transformer Model Performance

Dataset Metric Value (EURUSD)

Test RMSE 0.000320

Test MAE 0.000207

Validation RMSE 0.000405

Validation MAE 0.000277

Training RMSE 0.000400

Training MAE 0.000268

DROPOUT: 0.2

Epoch 01/20 -> Train Loss: -1.19325499, Val Loss: -1.59414840
New best validation loss: -1.59414840. Saving model...

Epoch 02/20 -> Train Loss: -2.14135868, Val Loss: -1.99962610
New best validation loss: -1.99962610. Saving model...

Epoch 03/20 -> Train Loss: -2.41637662, Val Loss: -2.37010575
New best validation loss: -2.37010575. Saving model...

Epoch 04/20 -> Train Loss: -2.54890556, Val Loss: -2.34356037
Validation loss did not improve. Patience: 1/10

Epoch 05/20 -> Train Loss: -2.61863458, Val Loss: -2.54329941
New best validation loss: -2.54329941. Saving model...

Epoch 06/20 -> Train Loss: -2.68047660, Val Loss: -2.17932529
Validation loss did not improve. Patience: 1/10

Epoch 07/20 -> Train Loss: -2.71625460, Val Loss: -2.57724358
New best validation loss: -2.57724358. Saving model...

Epoch 08/20 -> Train Loss: -2.75461274, Val Loss: -2.70489693
New best validation loss: -2.70489693. Saving model...

Epoch 09/20 -> Train Loss: -2.79114177, Val Loss: -2.60529278
Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -2.80865057, Val Loss: -2.71795976
New best validation loss: -2.71795976. Saving model...

Epoch 11/20 -> Train Loss: -2.84200477, Val Loss: -2.52086367
Validation loss did not improve. Patience: 1/10

Epoch 12/20 -> Train Loss: -2.86811929, Val Loss: -2.75316894
New best validation loss: -2.75316894. Saving model...

Epoch 13/20 -> Train Loss: -2.87644799, Val Loss: -2.86251446
New best validation loss: -2.86251446. Saving model...

Epoch 14/20 -> Train Loss: -2.89610735, Val Loss: -2.82280641
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -2.91163470, Val Loss: -2.85266695

Validation loss did not improve. Patience: 2/10

Epoch 16/20 -> Train Loss: -2.93229259, Val Loss: -2.72592718

Validation loss did not improve. Patience: 3/10

Epoch 17/20 -> Train Loss: -2.94098593, Val Loss: -2.79231819

Validation loss did not improve. Patience: 4/10

Epoch 18/20 -> Train Loss: -2.94783728, Val Loss: -2.92982869

New best validation loss: -2.92982869. Saving model...

Epoch 19/20 -> Train Loss: -2.95984276, Val Loss: -2.86058235

Validation loss did not improve. Patience: 1/10

Epoch 20/20 -> Train Loss: -2.96808456, Val Loss: -2.94427129

New best validation loss: -2.94427129

Final Transformer Model Performance		
Dataset	Metric	Value (EURUSD)
Test	RMSE	0.000503
Test	MAE	0.000401
Validation	RMSE	0.000716
Validation	MAE	0.000498
Training	RMSE	0.000415
Training	MAE	0.000284

DROPOUT: 0.3

Epoch 01/20 -> Train Loss: -1.11435746, Val Loss: -1.45170499
New best validation loss: -1.45170499. Saving model...

Epoch 02/20 -> Train Loss: -2.02066639, Val Loss: -1.83126043
New best validation loss: -1.83126043. Saving model...

Epoch 03/20 -> Train Loss: -2.31654629, Val Loss: -2.15474790
New best validation loss: -2.15474790. Saving model...

Epoch 04/20 -> Train Loss: -2.44832295, Val Loss: -2.29915429
New best validation loss: -2.29915429. Saving model...

Epoch 05/20 -> Train Loss: -2.51613078, Val Loss: -2.42867015
New best validation loss: -2.42867015. Saving model...

Epoch 06/20 -> Train Loss: -2.57449151, Val Loss: -2.20932350
Validation loss did not improve. Patience: 1/10

Epoch 07/20 -> Train Loss: -2.61606144, Val Loss: -2.49641295
New best validation loss: -2.49641295. Saving model...

Epoch 08/20 -> Train Loss: -2.64946251, Val Loss: -2.64031264
New best validation loss: -2.64031264. Saving model...

Epoch 09/20 -> Train Loss: -2.68607453, Val Loss: -2.60236787
Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -2.69528462, Val Loss: -2.72877364
New best validation loss: -2.72877364. Saving model...

Epoch 11/20 -> Train Loss: -2.72287390, Val Loss: -2.42659740
Validation loss did not improve. Patience: 1/10

Epoch 12/20 -> Train Loss: -2.75540312, Val Loss: -2.71711942
Validation loss did not improve. Patience: 2/10

Epoch 13/20 -> Train Loss: -2.75792165, Val Loss: -2.80767048
New best validation loss: -2.80767048. Saving model...

Epoch 14/20 -> Train Loss: -2.77669176, Val Loss: -2.75138762
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -2.79777672, Val Loss: -2.79019699

Validation loss did not improve. Patience: 2/10

Epoch 16/20 -> Train Loss: -2.81534322, Val Loss: -2.80056129

Validation loss did not improve. Patience: 3/10

Epoch 17/20 -> Train Loss: -2.82550183, Val Loss: -2.60367702

Validation loss did not improve. Patience: 4/10

Epoch 18/20 -> Train Loss: -2.83576179, Val Loss: -2.85979988

New best validation loss: -2.85979988. Saving model...

Epoch 19/20 -> Train Loss: -2.84640166, Val Loss: -2.41618766

Validation loss did not improve. Patience: 1/10

Epoch 20/20 -> Train Loss: -2.86078302, Val Loss: -2.90157366

New best validation loss: -2.90157366

Final Transformer Model Performance		
Dataset	Metric	Value (EURUSD)
Test	RMSE	0.000466
Test	MAE	0.000364
Validation	RMSE	0.000608
Validation	MAE	0.000429
Training	RMSE	0.000426
Training	MAE	0.000288